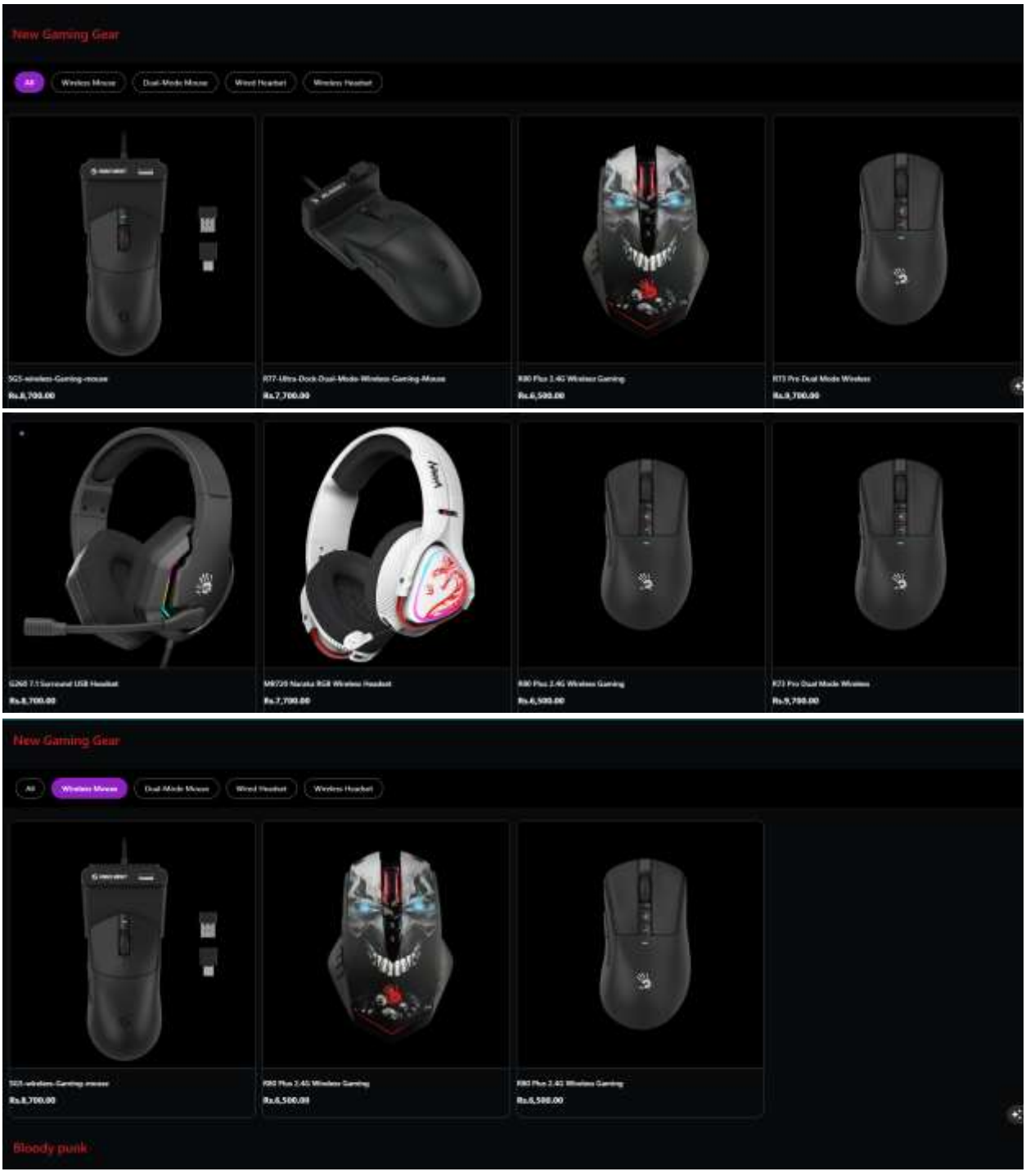
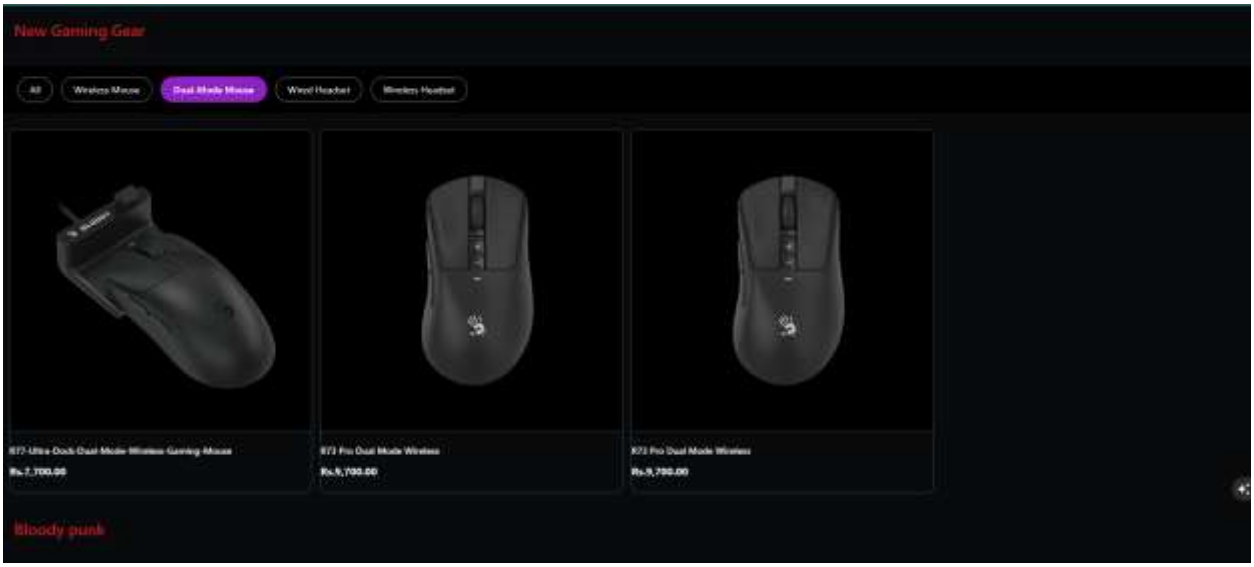
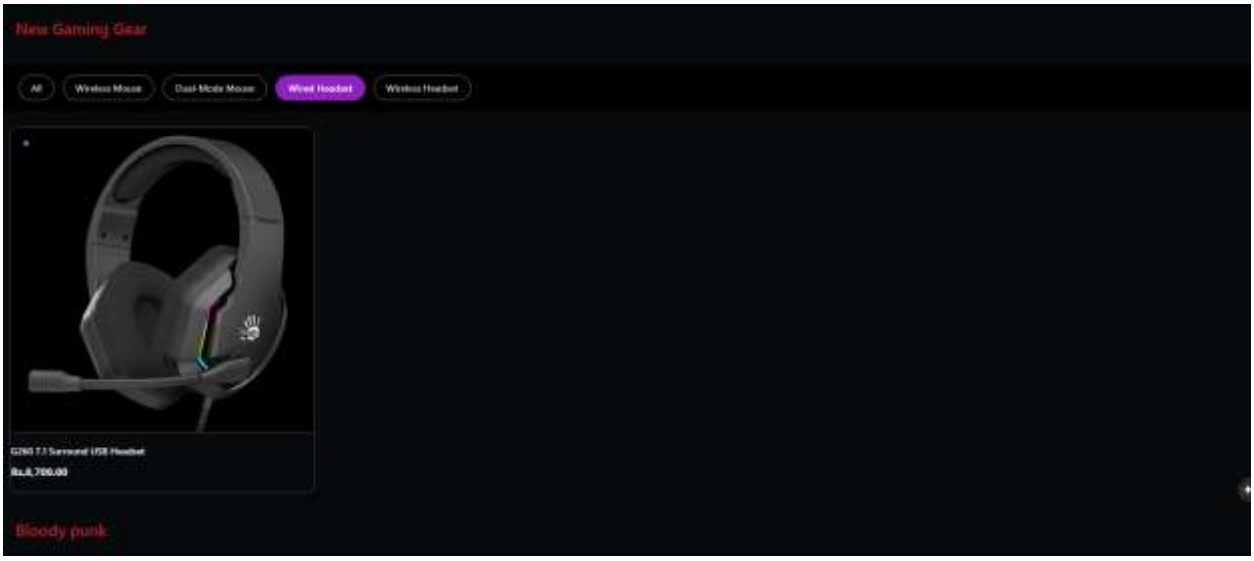
Hamza Khalid

FA24-BCS-040

Lab Task 1:



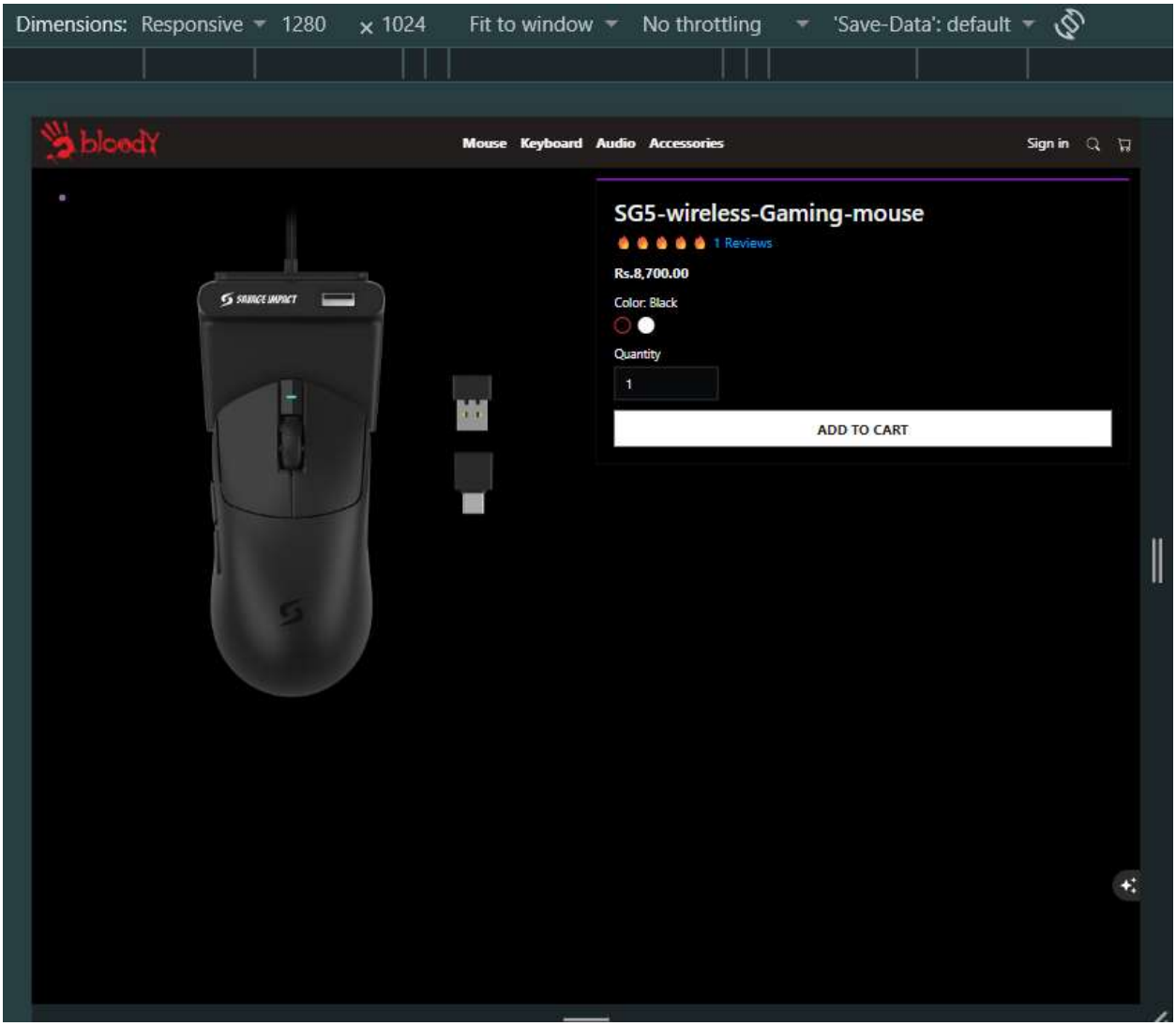

New Gaming Gear
All
Wireless Mouse
Dual Mode Mouse
Wired Headset
Wireless Headset
Bloody punk

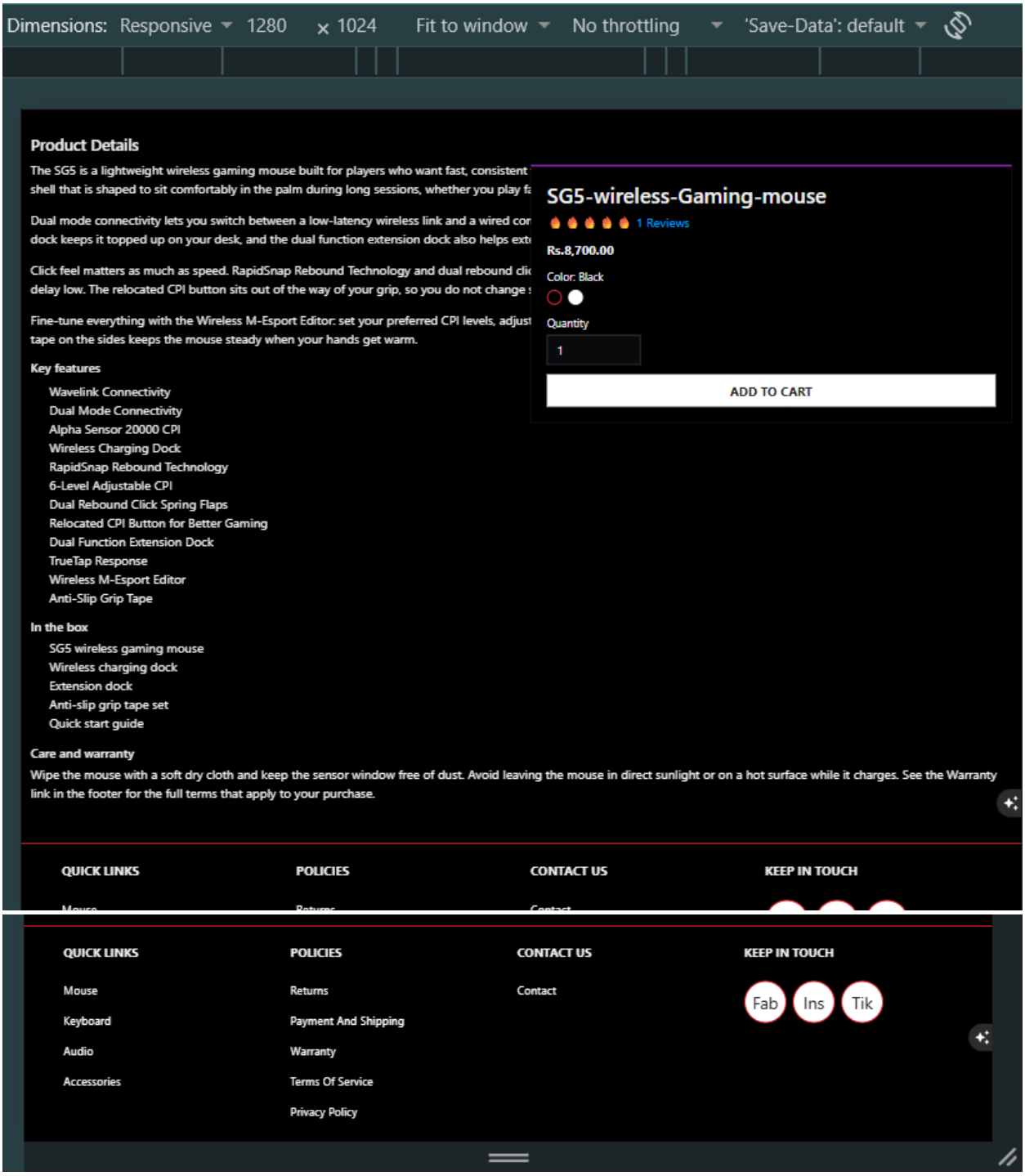

New Gaming Gear
All
Wireless Mouse
Dual Mode Mouse
Wired Headset
Wireless Headset
Bloody punk


New Gaming Gear
All
Wireless Mouse
Dual Mode Mouse
Wired Headset
Wireless Headset
Bloody punk

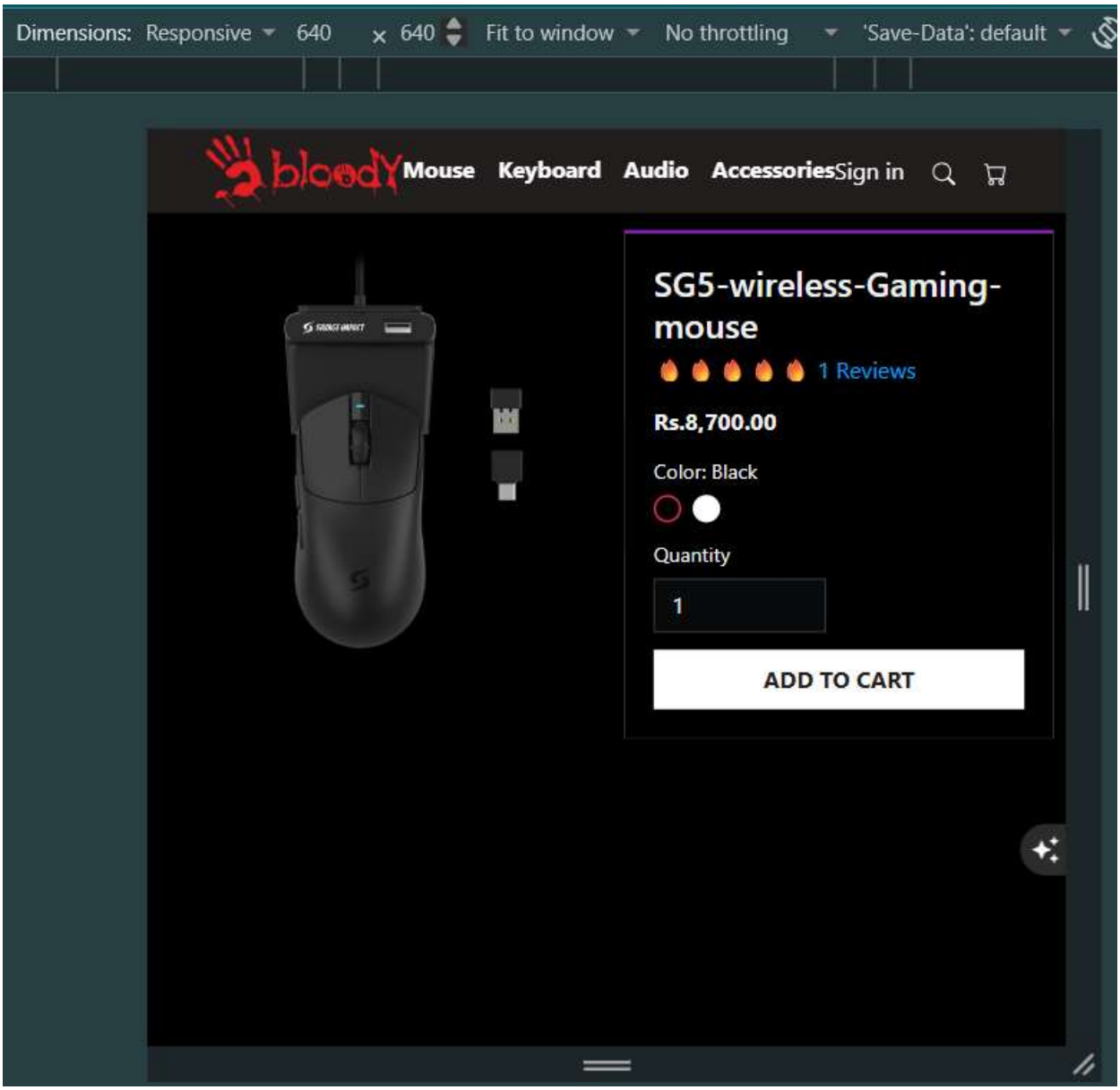
Lab Task 2:

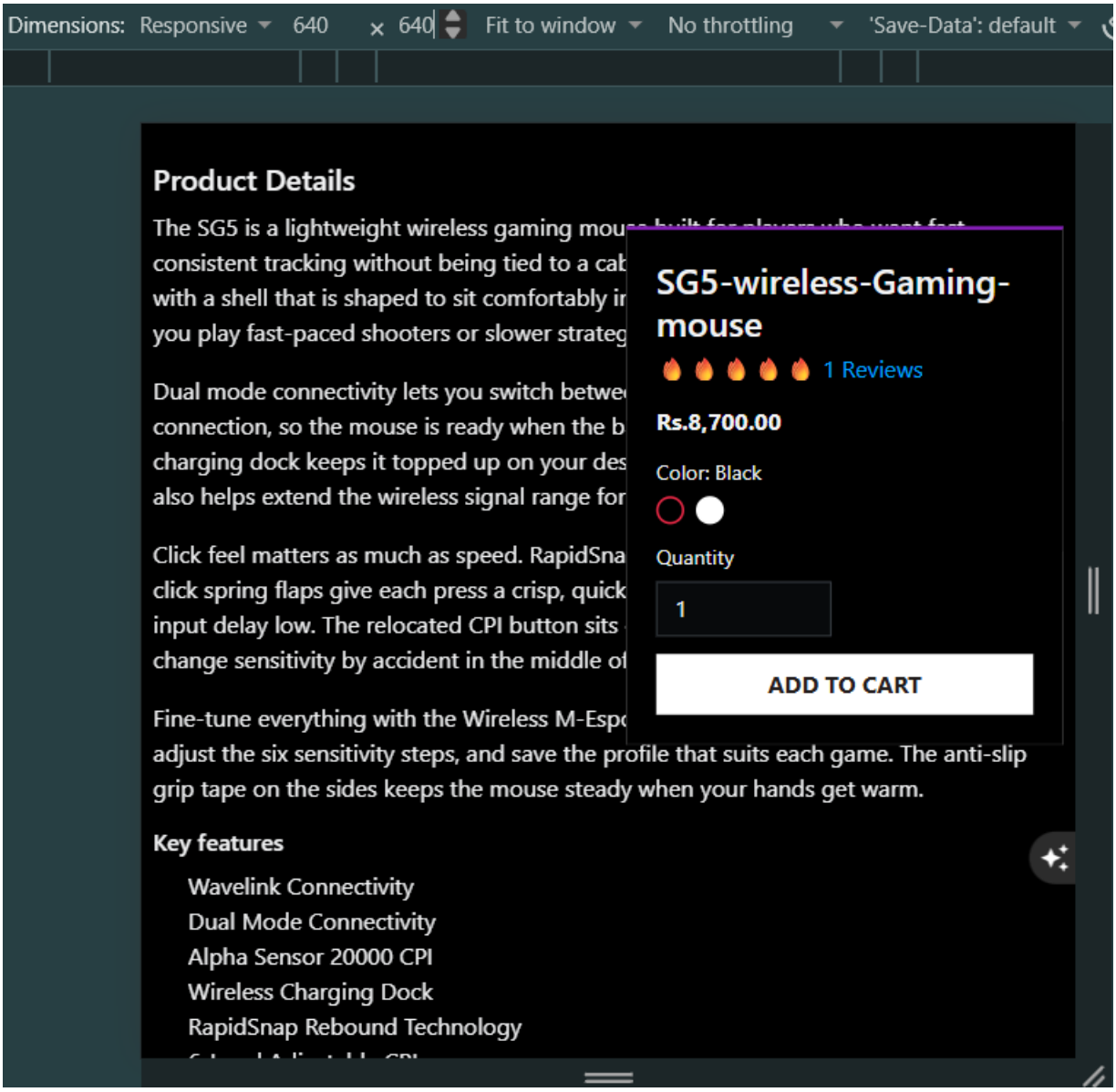
~1280 px:

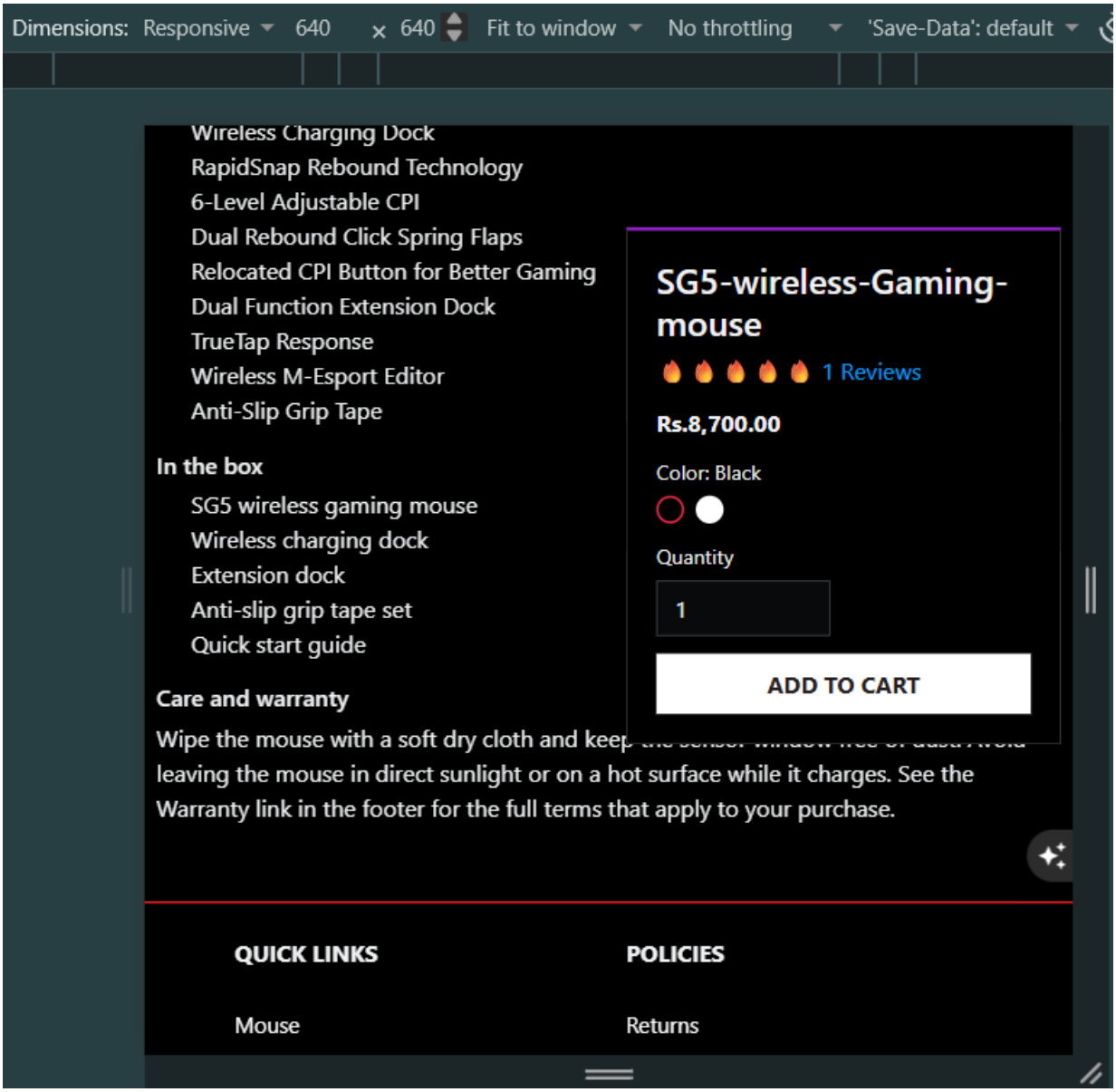


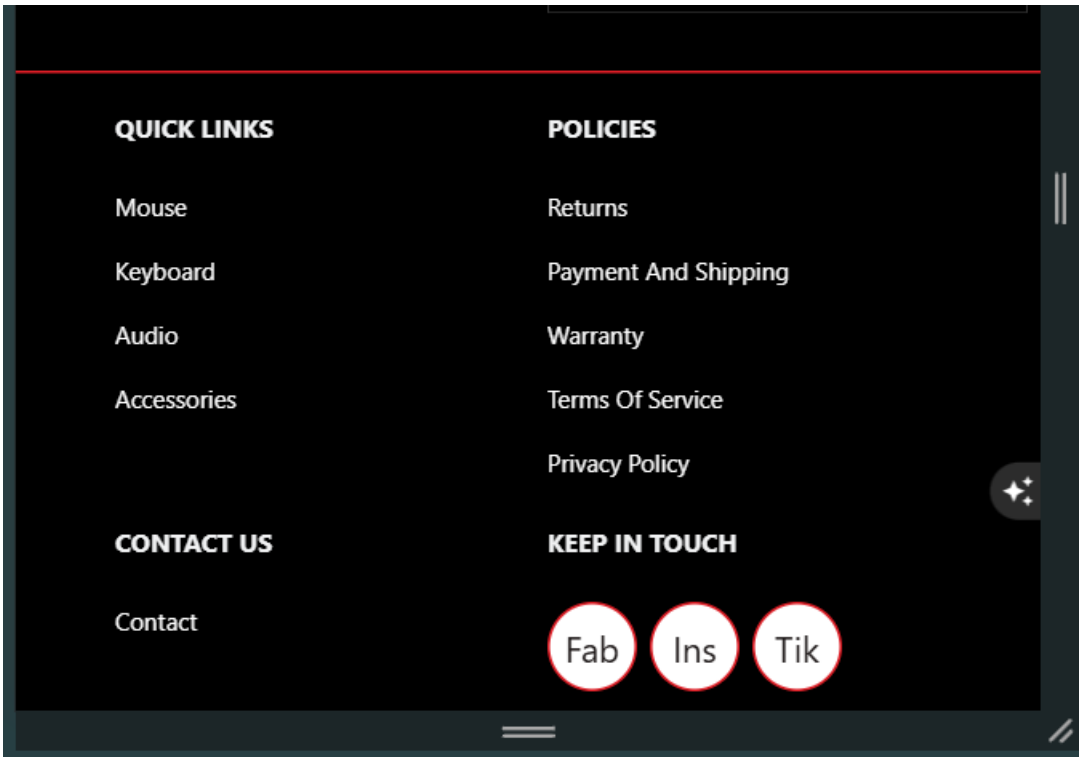

Dimensions: Responsive 1280 x 1024 Fit to window No throttling 'Save-Data': default
Product Details
The SG5 is a lightweight wireless gaming mouse built for players who want fast, consistent
shell that is shaped to sit comfortably in the palm during long sessions, whether you play fa
Dual mode connectivity lets you switch between a low-latency wireless link and a wired cor
dock keeps it topped up on your desk, and the dual function extension dock also helps ext
Click feel matters as much as speed. RapidSnap Rebound Technology and dual rebound cli
delay low. The relocated CPI button sits out of the way of your grip, so you do not change
Fine-tune everything with the Wireless M-Esport Editor: set your preferred CPI levels, adjust
tape on the sides keeps the mouse steady when your hands get warm.
SG5-wireless-Gaming-mouse
1 Reviews
Rs.8,700.00
Color: Black
Quantity
1
ADD TO CART
Key features
Wavelink Connectivity
Dual Mode Connectivity
Alpha Sensor 20000 CPI
Wireless Charging Dock
RapidSnap Rebound Technology
6-Level Adjustable CPI
Dual Rebound Click Spring Flaps
Relocated CPI Button for Better Gaming
Dual Function Extension Dock
TrueTap Response
Wireless M-Esport Editor
Anti-Slip Grip Tape
In the box
SG5 wireless gaming mouse
Wireless charging dock
Extension dock
Anti-slip grip tape set
Quick start guide
Care and warranty
Wipe the mouse with a soft dry cloth and keep the sensor window free of dust. Avoid leaving the mouse in direct sunlight or on a hot surface while it charges. See the Warranty link in the footer for the full terms that apply to your purchase.
QUICK LINKS
POLICIES
CONTACT US
KEEP IN TOUCH
QUICK LINKS
Mouse
Keyboard
Audio
Accessories
POLICIES
Returns
Payment And Shipping
Warranty
Terms Of Service
Privacy Policy
CONTACT US
Contact
KEEP IN TOUCH
Fab
Ins
Tik

~640px:

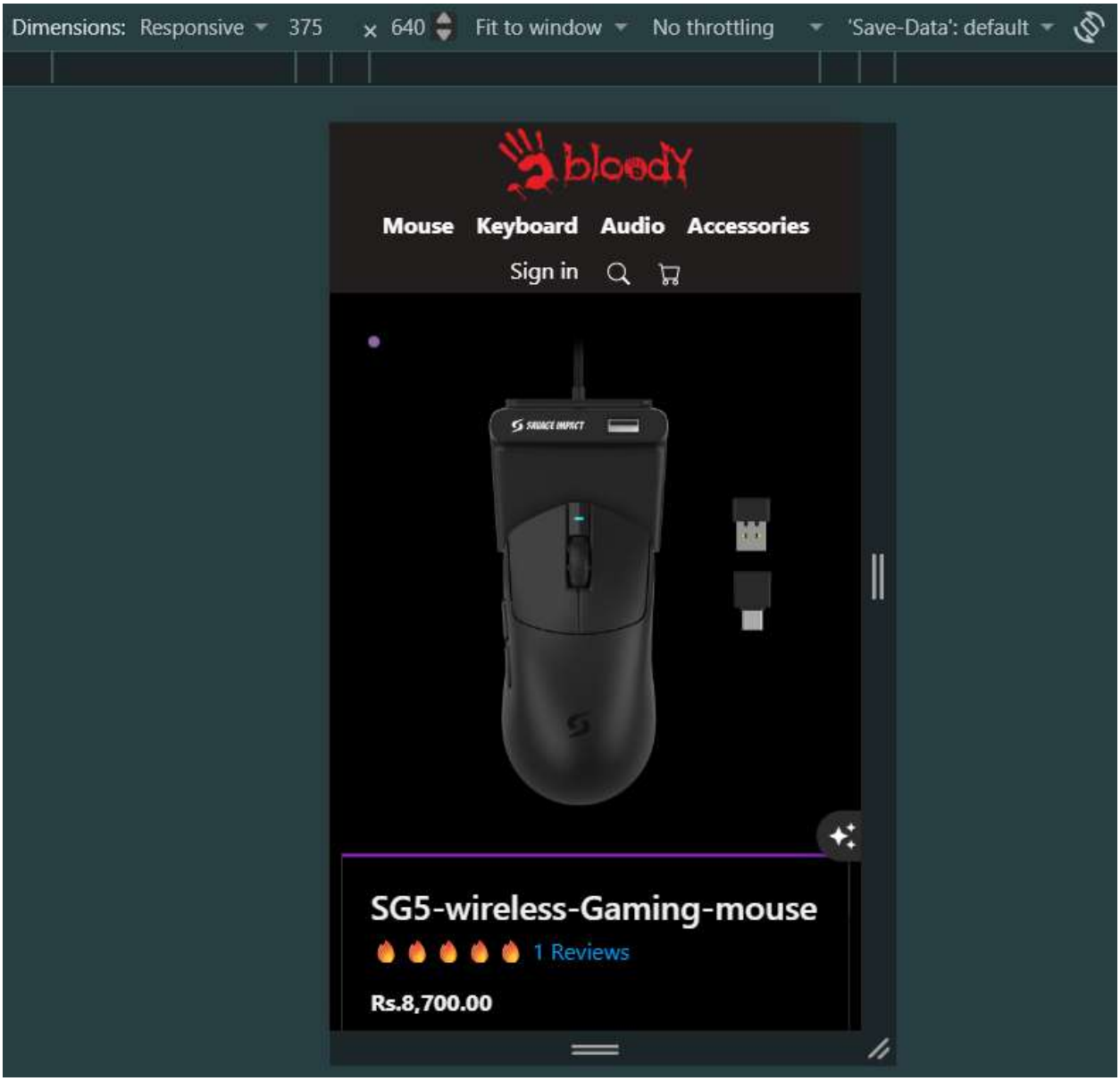

Dimensions: Responsive 640 x 640 Fit to window No throttling 'Save-Data': default
bloody
Mouse
Keyboard
Audio
Accessories
Sign in
SG5-wireless-Gaming-mouse
1 Reviews
Rs.8,700.00
Color: Black
Quantity
1
ADD TO CART

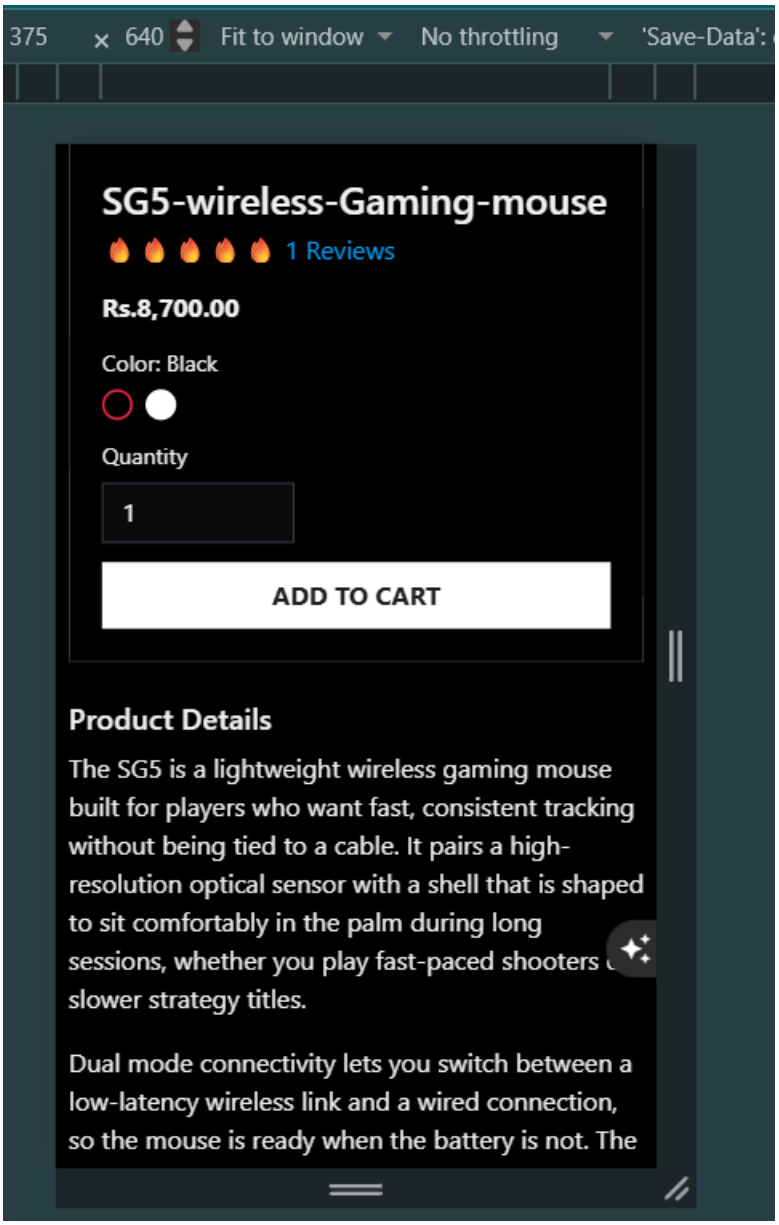

Dimensions: Responsive 640 x 640 Fit to window No throttling 'Save-Data': default
Product Details
The SG5 is a lightweight wireless gaming mou
consistent tracking without being tied to a cab
with a shell that is shaped to sit comfortably ir
you play fast-paced shooters or slower strateg
Dual mode connectivity lets you switch betwe
connection, so the mouse is ready when the b
charging dock keeps it topped up on your des
also helps extend the wireless signal range for
Click feel matters as much as speed. RapidSna
click spring flaps give each press a crisp, quick
input delay low. The relocated CPI button sits
change sensitivity by accident in the middle of
Fine-tune everything with the Wireless M-Espo
adjust the six sensitivity steps, and save the profile that suits each game. The anti-slip
grip tape on the sides keeps the mouse steady when your hands get warm.
Key features
Wavelink Connectivity
Dual Mode Connectivity
Alpha Sensor 20000 CPI
Wireless Charging Dock
RapidSnap Rebound Technology
SG5-wireless-Gaming-mouse
1 Reviews
Rs.8,700.00
Color: Black
Quantity
1
ADD TO CART

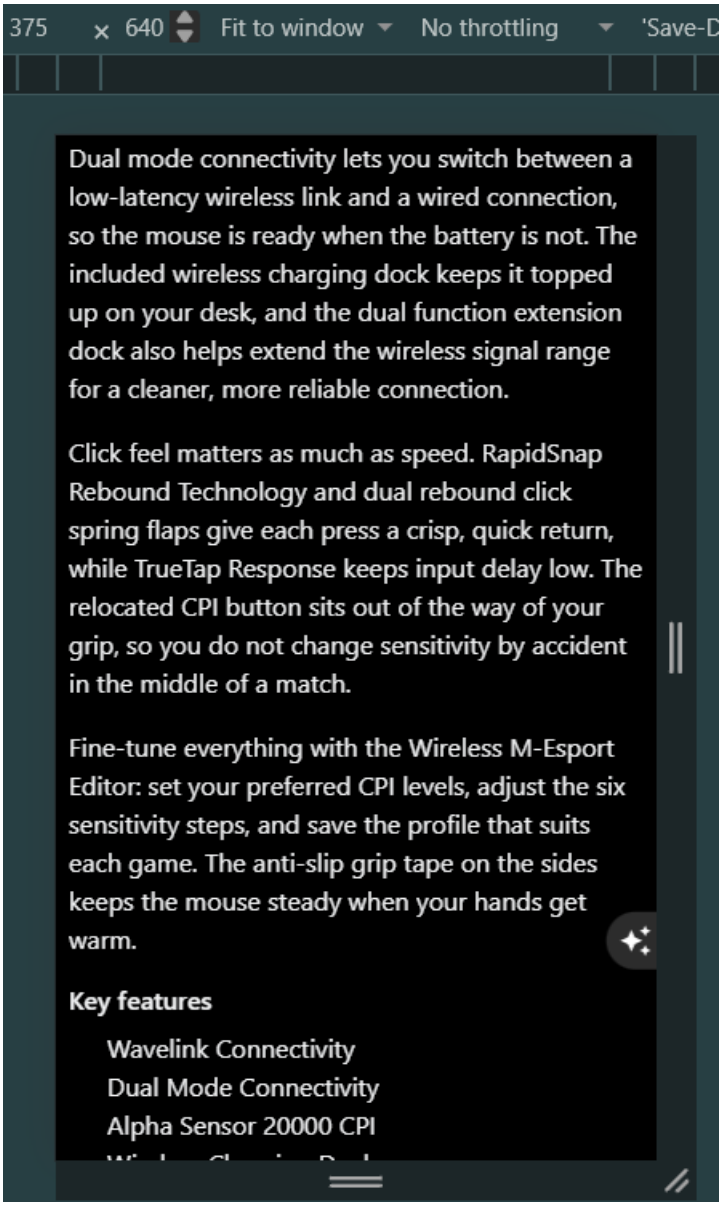

Dimensions: Responsive 640 x 640 Fit to window No throttling 'Save-Data': default
Wireless Charging Dock
RapidSnap Rebound Technology
6-Level Adjustable CPI
Dual Rebound Click Spring Flaps
Relocated CPI Button for Better Gaming
Dual Function Extension Dock
TrueTap Response
Wireless M-Esport Editor
Anti-Slip Grip Tape
In the box
SG5 wireless gaming mouse
Wireless charging dock
Extension dock
Anti-slip grip tape set
Quick start guide
Care and warranty
leaving the mouse in direct sunlight or on a hot surface while it charges. See the
Warranty link in the footer for the full terms that apply to your purchase.
SG5-wireless-Gaming-mouse
1 Reviews
Rs.8,700.00
Color: Black
Quantity
1
ADD TO CART
QUICK LINKS
POLICIES
Mouse
Returns

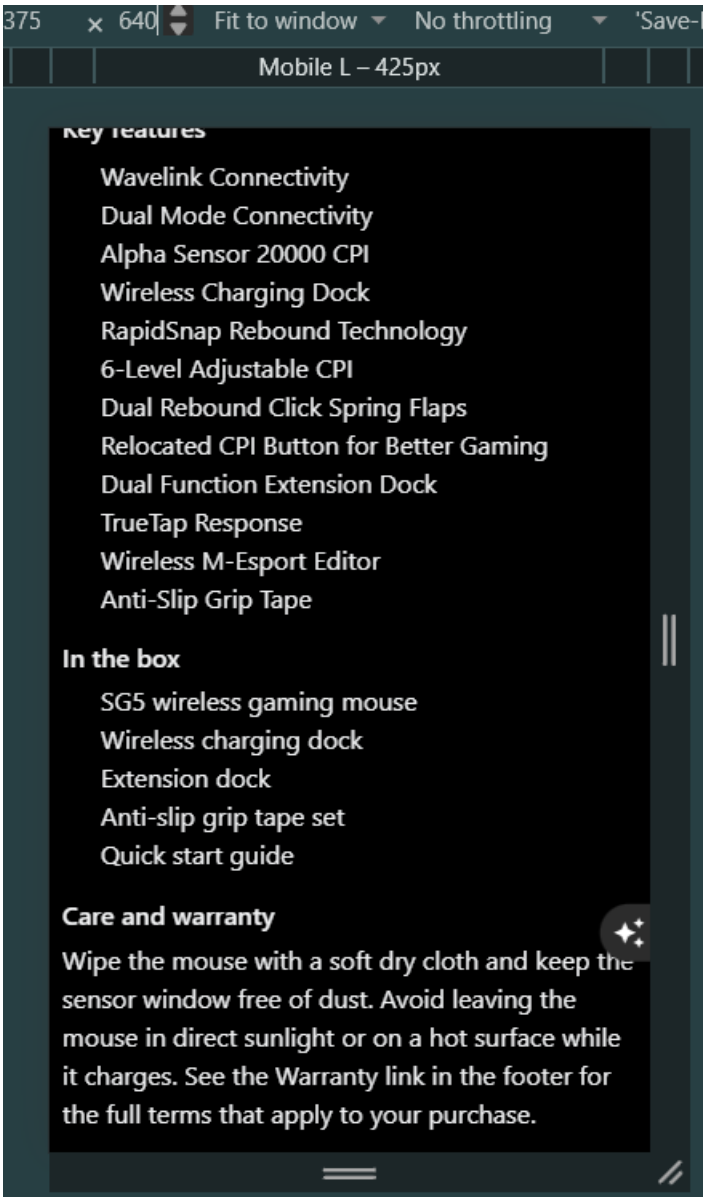

QUICK LINKS
Mouse
Keyboard
Audio
Accessories
POLICIES
Returns
Payment And Shipping
Warranty
Terms Of Service
Privacy Policy
CONTACT US
Contact
KEEP IN TOUCH
Fab
Ins
Tik

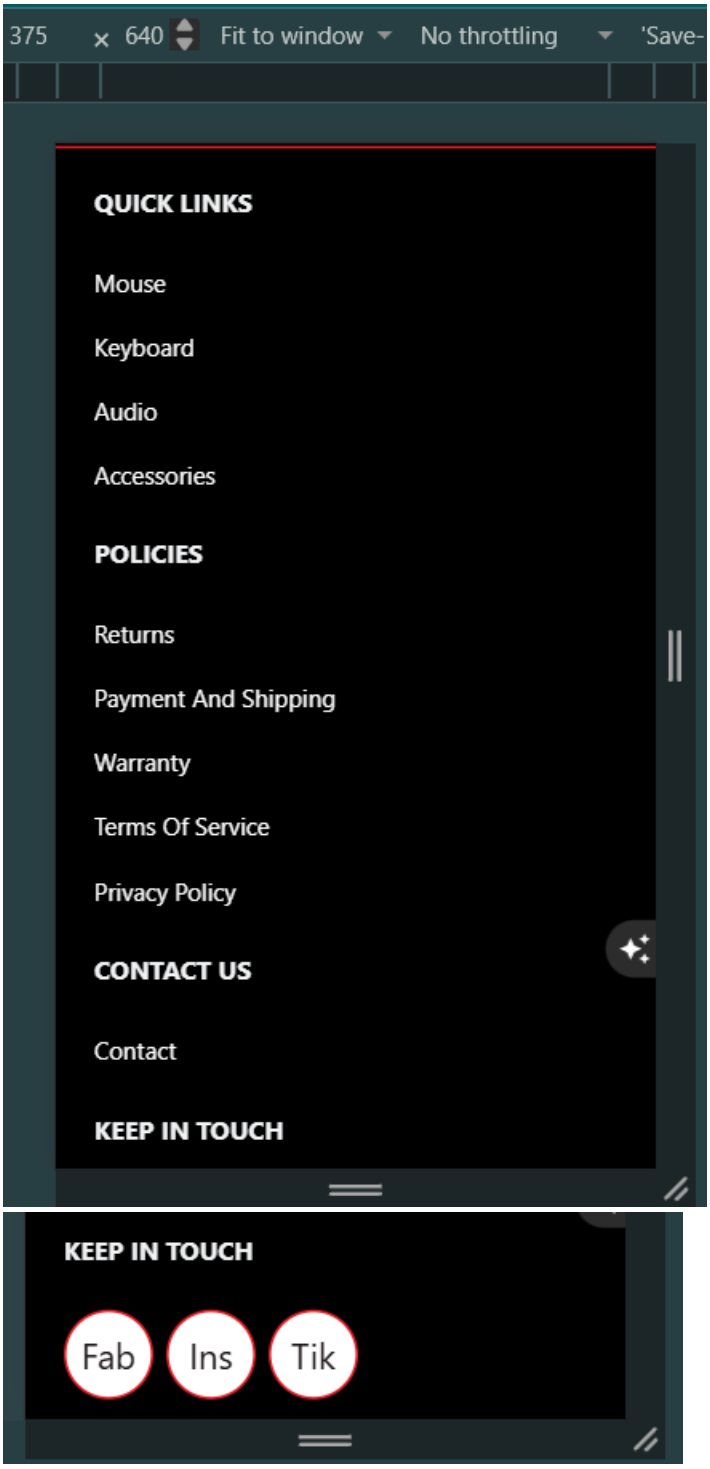
~375px:


Dimensions: Responsive 375 x 640 Fit to window No throttling 'Save-Data': default
bloody
Mouse Keyboard Audio Accessories
Sign in
SG5-wireless-Gaming-mouse
1 Reviews
Rs.8,700.00


375 × 640 Fit to window No throttling 'Save-Data':
SG5-wireless-Gaming-mouse
1 Reviews
Rs.8,700.00
Color: Black
Quantity
1
ADD TO CART
Product Details
The SG5 is a lightweight wireless gaming mouse built for players who want fast, consistent tracking without being tied to a cable. It pairs a high-resolution optical sensor with a shell that is shaped to sit comfortably in the palm during long sessions, whether you play fast-paced shooters
slower strategy titles.
Dual mode connectivity lets you switch between a low-latency wireless link and a wired connection, so the mouse is ready when the battery is not. The

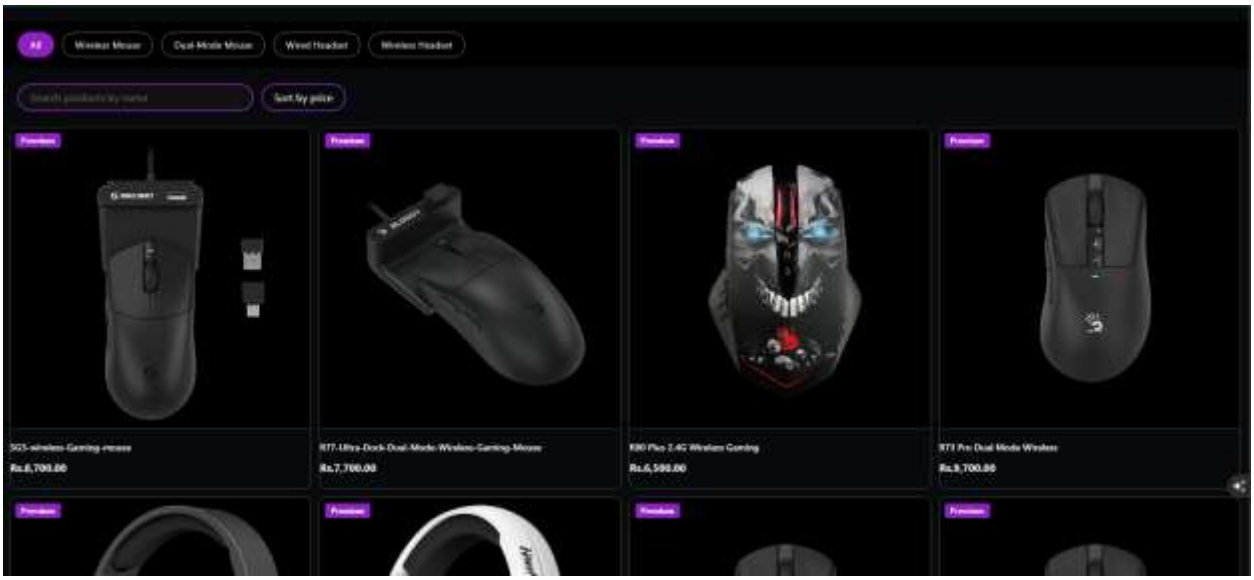

375 × 640 Fit to window No throttling 'Save-D
Dual mode connectivity lets you switch between a low-latency wireless link and a wired connection, so the mouse is ready when the battery is not. The included wireless charging dock keeps it topped up on your desk, and the dual function extension dock also helps extend the wireless signal range for a cleaner, more reliable connection.
Click feel matters as much as speed. RapidSnap Rebound Technology and dual rebound click spring flaps give each press a crisp, quick return, while TrueTap Response keeps input delay low. The relocated CPI button sits out of the way of your grip, so you do not change sensitivity by accident in the middle of a match.
Fine-tune everything with the Wireless M-Esport Editor: set your preferred CPI levels, adjust the six sensitivity steps, and save the profile that suits each game. The anti-slip grip tape on the sides keeps the mouse steady when your hands get warm.
Key features
Wavelink Connectivity
Dual Mode Connectivity
Alpha Sensor 20000 CPI


375 × 640
Fit to window
No throttling
Mobile L – 425px
Wavelink Connectivity
Dual Mode Connectivity
Alpha Sensor 20000 CPI
Wireless Charging Dock
RapidSnap Rebound Technology
6-Level Adjustable CPI
Dual Rebound Click Spring Flaps
Relocated CPI Button for Better Gaming
Dual Function Extension Dock
TrueTap Response
Wireless M-Esport Editor
Anti-Slip Grip Tape
In the box
SG5 wireless gaming mouse
Wireless charging dock
Extension dock
Anti-slip grip tape set
Quick start guide
Care and warranty
Wipe the mouse with a soft dry cloth and keep the sensor window free of dust. Avoid leaving the mouse in direct sunlight or on a hot surface while it charges. See the Warranty link in the footer for the full terms that apply to your purchase.

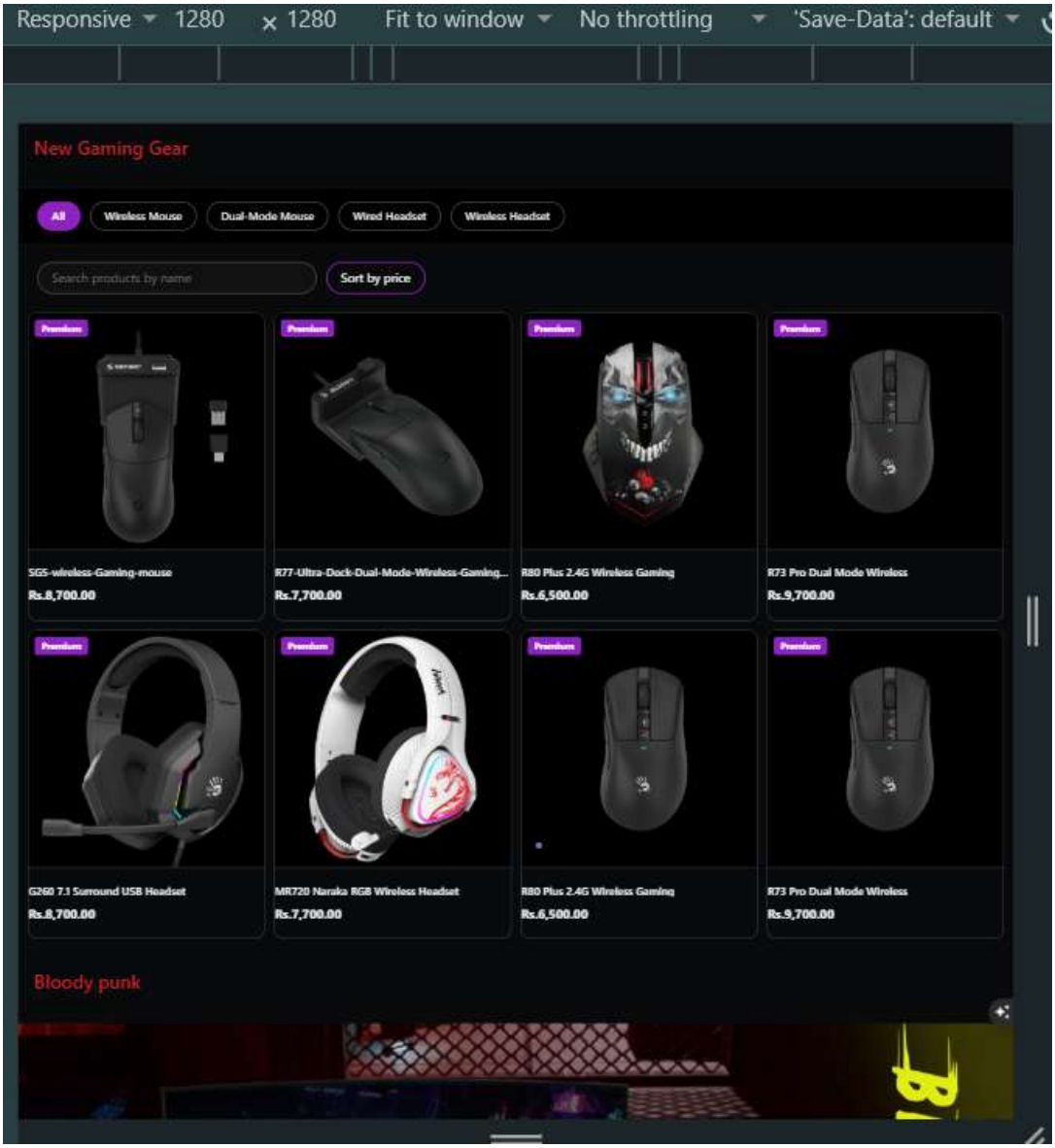


Task 3:

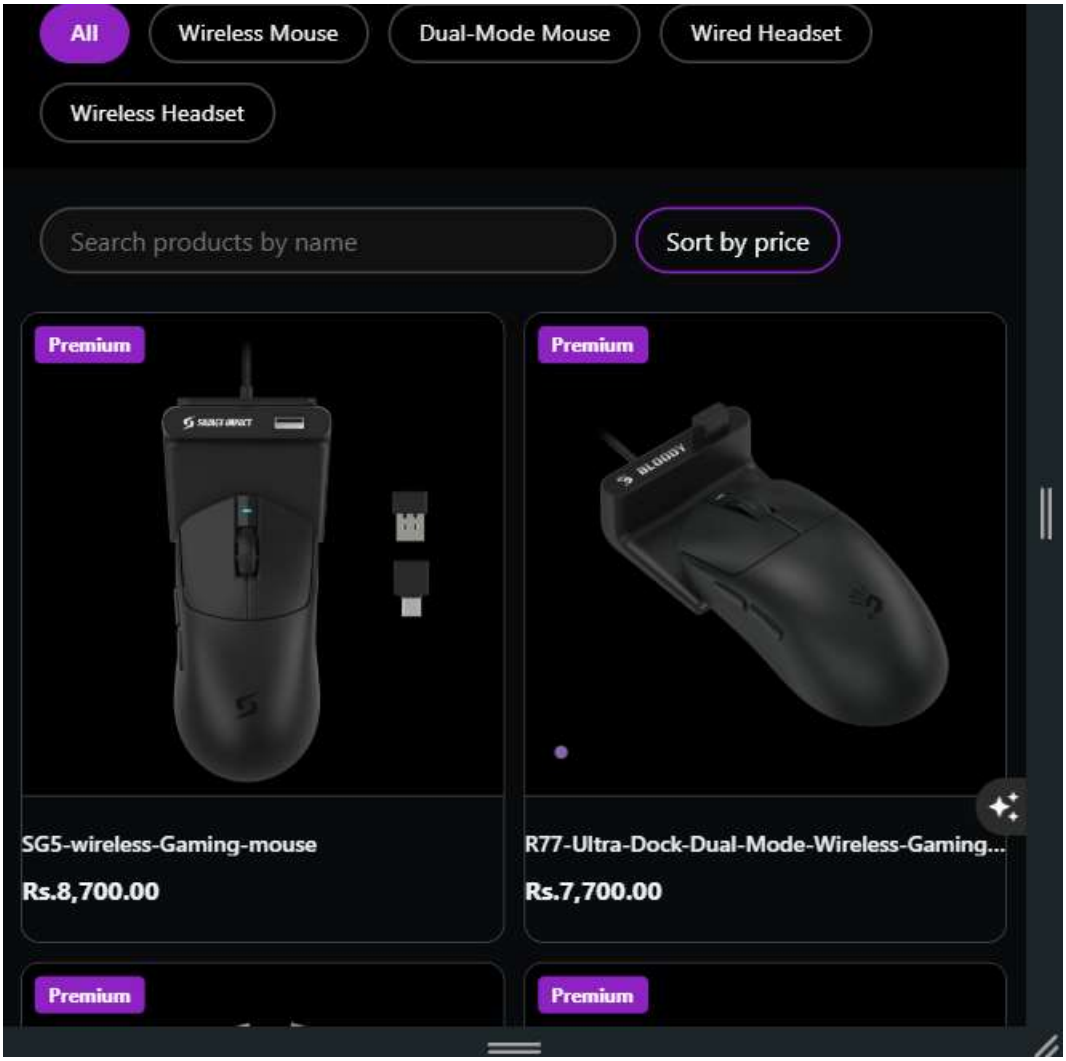
Desktop view:


All
Wireless Mouse
Dual-Mode Mouse
Wired Headset
Wireless Headset
Sort by price
Premium
Premium
Premium
Premium
Rs.6,700.00
Rs.7,700.00
Rs.6,500.00
Rs.3,700.00
Premium
Premium
Premium
Premium

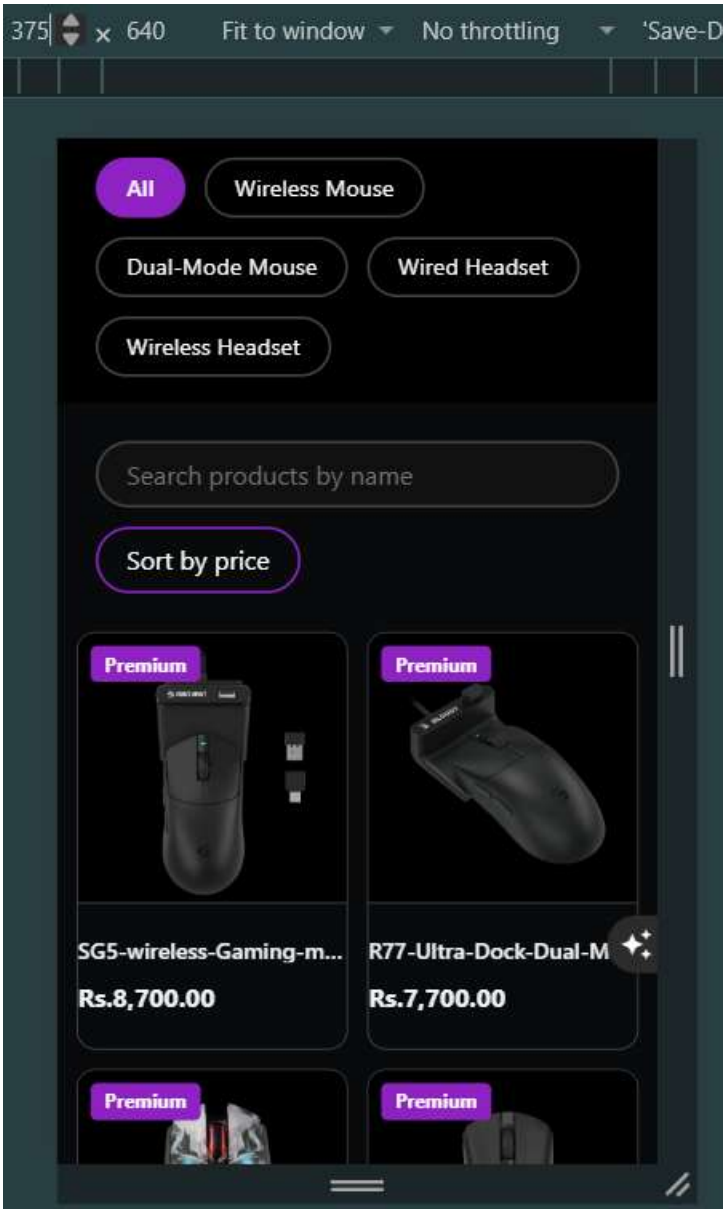
~1280px:


Responsive 1280 × 1280 Fit to window No throttling 'Save-Data': default
New Gaming Gear
All
Wireless Mouse
Dual-Mode Mouse
Wired Headset
Wireless Headset
Search products by name
Sort by price
Premium
SGS-wireless-Gaming-mouse
Rs.8,700.00
R77-Ultra-Dock-Dual-Mode-Wireless-Gaming...
Rs.7,700.00
R80 Plus 2.4G Wireless Gaming
Rs.6,500.00
R73 Pro Dual Mode Wireless
Rs.9,700.00
G260 7.1 Surround USB Headset
Rs.8,700.00
MR720 Naraka RGB Wireless Headset
Rs.7,700.00
R80 Plus 2.4G Wireless Gaming
Rs.6,500.00
R73 Pro Dual Mode Wireless
Rs.9,700.00
Bloody punk

~640 px:

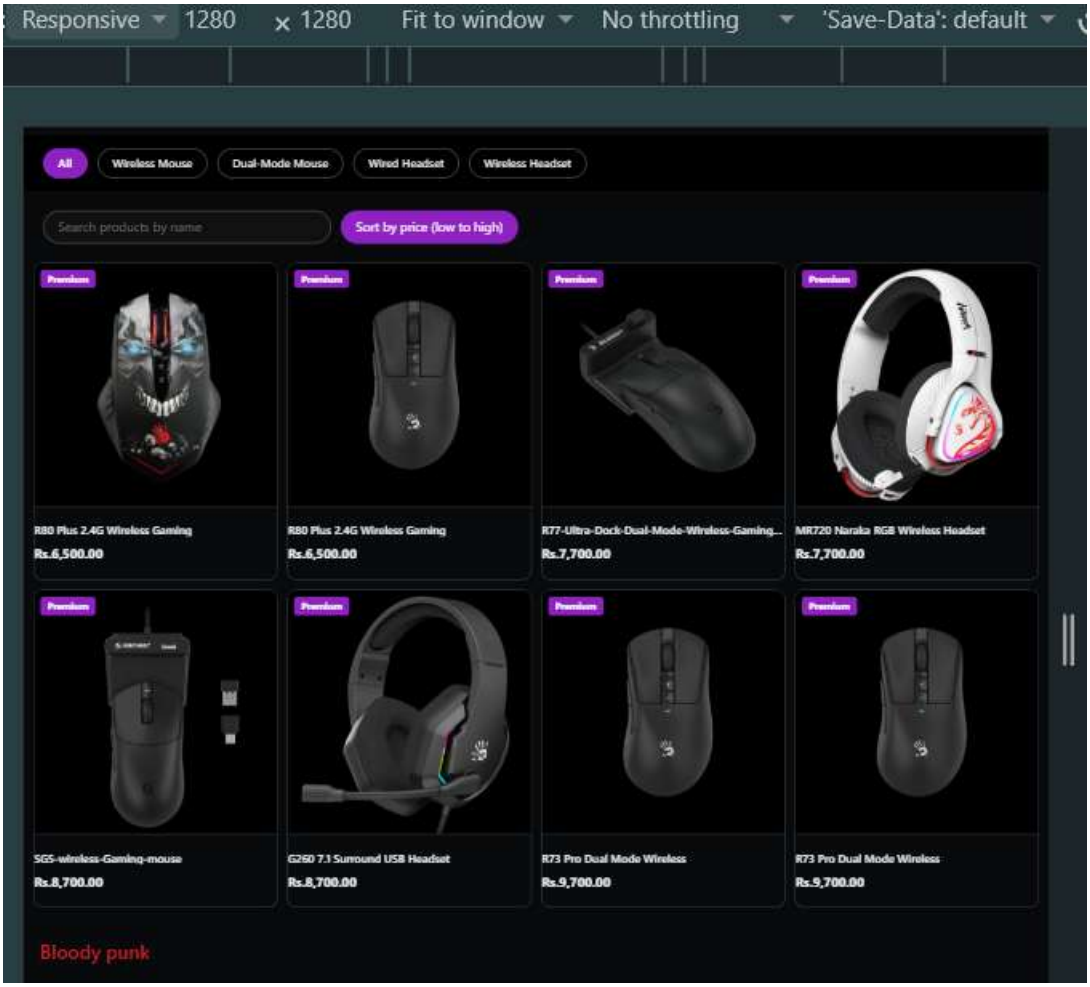

All
Wireless Mouse
Dual-Mode Mouse
Wired Headset
Wireless Headset
Search products by name
Sort by price
Premium
Premium
SG5-wireless-Gaming-mouse
Rs.8,700.00
R77-Ultra-Dock-Dual-Mode-Wireless-Gaming...
Rs.7,700.00
Premium
Premium

~375 px:

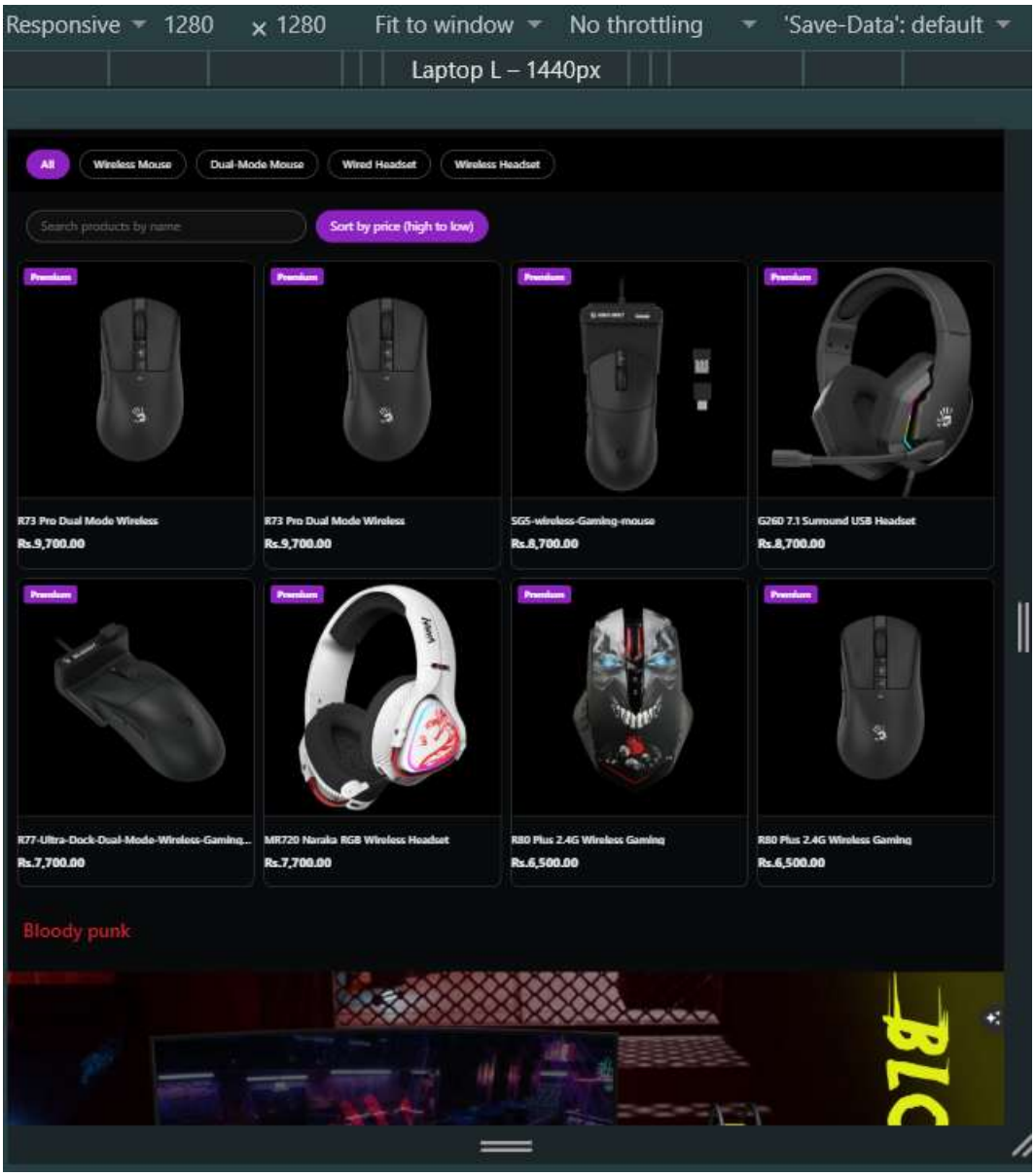


Testing Functionality:

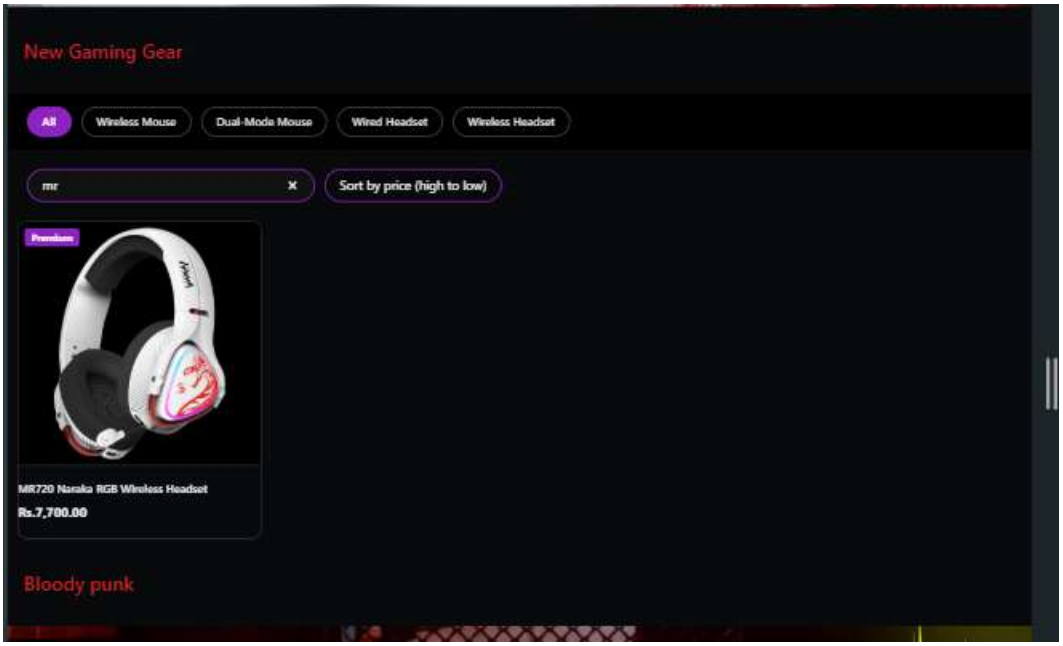
Sorting from low to high price:



Sorting from high to low price:



Searching by name:



Task 1:

Code added:

```
<!-- Task 1: CSS-only category filter (radios must be siblings that come BEFORE the grid) -->
<div class="shop">
<input type="radio" name="category" id="f-all" class="filter-radio" checked>
<input type="radio" name="category" id="f-wireless-mouse" class="filter-radio">
<input type="radio" name="category" id="f-dual-mode-mouse" class="filter-radio">
<input type="radio" name="category" id="f-wired-headset" class="filter-radio">
<input type="radio" name="category" id="f-wireless-headset" class="filter-radio">

<div class="filter-bar" role="group" aria-label="Filter products by category">
<label for="f-all">All</label>
<label for="f-wireless-mouse">Wireless Mouse</label>
<label for="f-dual-mode-mouse">Dual-Mode Mouse</label>
<label for="f-wired-headset">Wired Headset</label>
<label for="f-wireless-headset">Wireless Headset</label>
</div>
```

Style.css code added:

```
/* ===== Task 1: CSS-only category filter (accent hue H = 280) ===== */
:root {
  --accent: hsl(280, 70%, 45%);
}

/* Radios stay in the DOM (keyboard + screen-reader accessible) but are visually hidden */
.shop {
  position: relative;
}
.filter-radio {
  position: absolute;
  opacity: 0;
  width: 1px;
  height: 1px;
  pointer-events: none;
}

/* Filter bar */
.filter-bar {
  display: flex;
  flex-wrap: wrap;
  gap: 0.75rem;
  padding: 1rem 1.5rem;
  margin-bottom: 1.5rem;
  background: #000;
}
.filter-bar label {
  cursor: pointer;
  color: #fff;
  border: 2px solid #444;
  border-radius: 999px;
  padding: 0.4rem 1.1rem;
  font-size: 0.9rem;
  transition: background-color 0.15s, border-color 0.15s;
}
.filter-bar label:hover {
  border-color: var(--accent);
}

/* Selected filter looks selected (accent colour) */
#f-all:checked ~ .filter-bar label[for="f-all"],
#f-wireless-mouse:checked ~ .filter-bar label[for="f-wireless-mouse"],
```

```
#f-dual-mode-mouse:checked ~ .filter-bar label[for="f-dual-mode-mouse"],
#f-wired-headset:checked ~ .filter-bar label[for="f-wired-headset"],
#f-wireless-headset:checked ~ .filter-bar label[for="f-wireless-headset"] {
  background: var(--accent);
  border-color: var(--accent);
  font-weight: 600;
}

/* Keyboard focus ring on the label of the focused radio */
#f-all:focus-visible ~ .filter-bar label[for="f-all"],
#f-wireless-mouse:focus-visible ~ .filter-bar label[for="f-wireless-mouse"],
#f-dual-mode-mouse:focus-visible ~ .filter-bar label[for="f-dual-mode-mouse"],
#f-wired-headset:focus-visible ~ .filter-bar label[for="f-wired-headset"],
#f-wireless-headset:focus-visible ~ .filter-bar label[for="f-wireless-headset"] {
  outline: 3px solid #fff;
  outline-offset: 2px;
}

/* Filtering: hide every card whose data-category does not match.
   display:none removes the card from the grid flow, so no empty gaps are left. */
#f-wireless-mouse:checked ~ .product-section .product-item:not([data-category="wireless-
mouse"]),
#f-dual-mode-mouse:checked ~ .product-section .product-item:not([data-category="dual-
mode-mouse"]),
#f-wired-headset:checked ~ .product-section .product-item:not([data-category="wired-
headset"]),
#f-wireless-headset:checked ~ .product-section .product-item:not([data-category="wireless-
headset"]) {
  display: none;
}
```

Task 2:

products.html

A few changes and added more description to verify dticky details:

```
<!-- Task 2: CSS Grid with named grid-template-areas (layout lives in style.css) -->
<section class="container-fluid px-3 lg:px-4 py-4">
<div class="product-layout">

<!-- area: image -->
<div class="product-image">
<img
```

```
id="product-image"
src="../Material_For_Website/SG5-wireless-Gaming-mouse.png"
alt="SG5 wireless gaming mouse"
class="img-fluid d-block mx-auto"
/>
</div>

<!-- area: details (sticky from 640px up) -->
<div class="product-details">
<h1 id="product-name" class="h3 text-danger">SG5-wireless-Gaming-mouse</h1>

<p class="mb-4">
<span aria-label="5 out of 5">🔥 🔥 🔥 🔥 🔥 </span>
<a href="#" class="text-danger text-decoration-none">1 Reviews</a>
</p>

<p id="product-price" class="fs-4 fw-bold mb-4">Rs.8,700.00</p>

<p class="small mb-2">Color: <span id="color" class="text-danger">Black</span></p>

<div class="d-flex gap-3 mb-4">
<span class="rounded-circle border border-2 border-danger p-3" title="Red"></span>
<span class="rounded-circle border border-2 border-white bg-white p-3"
title="Black"></span>
</div>

<div class="mb-4">
<label for="quantity" class="small d-block mb-2">Quantity</label>
<input type="number" id="quantity" name="quantity" class="form-control rounded-0
quantity-input" min="1" max="10" value="1">
</div>

<button type="button" class="btn bg-white color-black rounded-0 text-uppercase w-100 py-
3">
Add to cart
</button>
</div>

<!-- area: description (full width underneath from 640px up) -->
<div class="product-description">
<h2 class="h4 text-danger mb-3">Product Details</h2>

<p id="product-description">
The SG5 is a lightweight wireless gaming mouse built for players who want fast, consistent
```

tracking without being tied to a cable. It pairs a high-resolution optical sensor with a shell that is shaped to sit comfortably in the palm during long sessions, whether you play fast-paced shooters or slower strategy titles.

Dual mode connectivity lets you switch between a low-latency wireless link and a wired connection, so the mouse is ready when the battery is not. The included wireless charging dock keeps it topped up on your desk, and the dual function extension dock also helps extend the wireless signal range for a cleaner, more reliable connection.

Click feel matters as much as speed. RapidSnap Rebound Technology and dual rebound click spring flaps give each press a crisp, quick return, while TrueTap Response keeps input delay low. The relocated CPI button sits out of the way of your grip, so you do not change sensitivity by accident in the middle of a match.

Fine-tune everything with the Wireless M-Esport Editor: set your preferred CPI levels, adjust the six sensitivity steps, and save the profile that suits each game. The anti-slip grip tape on the sides keeps the mouse steady when your hands get warm.

<h3 class="h5 mt-4">Key features</h3>
<ul id="product-features" class="mt-3">
Wavelink Connectivity
Dual Mode Connectivity
Alpha Sensor 20000 CPI
Wireless Charging Dock
RapidSnap Rebound Technology
6-Level Adjustable CPI
Dual Rebound Click Spring Flaps
Relocated CPI Button for Better Gaming
Dual Function Extension Dock
TrueTap Response
Wireless M-Esport Editor
Anti-Slip Grip Tape

<h3 class="h5 mt-4">In the box</h3>

SG5 wireless gaming mouse
Wireless charging dock
Extension dock
Anti-slip grip tape set

```
<li>Quick start guide</li>
</ul>

<h3 class="h5 mt-4">Care and warranty</h3>
<p>
Wipe the mouse with a soft dry cloth and keep the sensor window free of dust. Avoid
leaving the mouse in direct sunlight or on a hot surface while it charges. See the
Warranty link in the footer for the full terms that apply to your purchase.
</p>
</div>

</div>
</section>
```

Style.css added

```
/* ===== Task 2: product page layout (BP = 640px, sticky top = 70px) ===== */

/* Mobile first (below 640px): one column, stacked image -> details -> description */
.product-layout {
  display: grid;
  grid-template-columns: 1fr;
  grid-template-areas:
    "image"
    "details"
    "description";
  gap: 1.5rem;
}

.product-image       { grid-area: image; }
.product-details     { grid-area: details; }
.product-description { grid-area: description; }

.product-details {
  background: #000;
  border: 1px solid #333;
  border-top: 3px solid var(--accent);
  padding: 1.25rem;
}

.quantity-input {
  max-width: 120px;
```

```
}

/* 640px and wider: image + details side by side, description full width underneath */
@media (min-width: 640px) {
  .product-layout {
    grid-template-columns: 1fr 1fr;
    grid-template-areas:
      "image      details"
      "description description";
  }

  /* The image area is made tall enough that the sticky box has room to travel inside it */
  .product-image {
    min-height: 140vh;
  }

  /* Sticky details box: stays 70px below the top of the viewport.
     align-self: start stops the grid from stretching it to the full row height
     (a stretched box has no room left to move, so sticky would do nothing). */
  .product-details {
    position: sticky;
    top: 70px;
    align-self: start;
  }
}
```

:

Task 3:

Added script.js:

```
// Task 3: live search, price sort and Premium label (plain JavaScript, works on the existing
cards)
(function () {
  var N = 40;                  // last two digits of FA24-BCS-040
  var PREMIUM_LIMIT = N * 100;    // products above Rs.4,000 get a Premium label

  var grid = document.querySelector('.product-grid');
  var searchInput = document.getElementById('search-input');
  var sortBtn = document.getElementById('sort-price');
  var sortDir = document.getElementById('sort-dir');
  var noResults = document.getElementById('no-results');
  var radios = document.querySelectorAll('.filter-radio');
```

```
// Always read the cards that are already in the page (their order changes after sorting)
function getCards() {
  return Array.prototype.slice.call(grid.querySelectorAll('.product-item'));
}

function getName(card) {
  return card.querySelector('.card-title').textContent.trim().toLowerCase();
}

// "Rs.8,700.00" -> 8700
function getPrice(card) {
  var text = card.querySelector('.card-text').textContent;
  var match = text.match(/\d[\d,]*(?:\.\d+)?/);
  return match ? parseFloat(match[0].replace(/,/g, '')) : NaN;
}

// A card is visible only if it passes BOTH filters:
//  - the Task 1 category filter (CSS hides non-matching cards with display: none)
//  - the search filter (this script adds .search-hidden to non-matching cards)
// So we just count the cards that are not display:none after both have been applied.
function updateEmptyState() {
  var anyVisible = getCards().some(function (card) {
    return getComputedStyle(card).display !== 'none';
  });
  grid.hidden = !anyVisible;
  noResults.hidden = anyVisible;
}

function applySearch() {
  var query = searchInput.value.trim().toLowerCase();
  getCards().forEach(function (card) {
    card.classList.toggle('search-hidden', !getName(card).includes(query));
  });
  updateEmptyState();
}

// Sort by price: first click low -> high, next click high -> low, and so on
var ascending = true;
function sortByPrice() {
  var sorted = getCards().sort(function (a, b) {
    return ascending ? getPrice(a) - getPrice(b) : getPrice(b) - getPrice(a);
  });
  sorted.forEach(function (card) {
```

```
    grid.appendChild(card);     // appendChild moves the existing element, nothing is rebuilt
   });
   sortDir.textContent = ascending ? '(low to high)' : '(high to low)';
   ascending = !ascending;
  }

  // Premium label, added once when the page loads
  function addPremiumLabels() {
   getCards().forEach(function (card) {
    if (getPrice(card) > PREMIUM_LIMIT) {
     var label = document.createElement('span');
     label.className = 'premium-badge';
     label.textContent = 'Premium';
     card.querySelector('.card').appendChild(label);
    }
   });
  }

  searchInput.addEventListener('input', applySearch);
  sortBtn.addEventListener('click', sortByPrice);
  radios.forEach(function (radio) {
   radio.addEventListener('change', updateEmptyState);   // keep "No products found" in sync
with Task 1
  });

  addPremiumLabels();
  applySearch();
})();
```

Added index.html code:

```
<!-- Task 3: live search + price sort (behaviour in script.js) -->
<div class="tools-bar">
<label for="search-input" class="visually-hidden">Search products by name</label>
<input type="search" id="search-input" class="search-input" placeholder="Search products
by name" autocomplete="off">
<button type="button" id="sort-price" class="sort-btn">Sort by price <span id="sort-
dir"></span></button>
</div>
```

Added style.css:

```
/* ===== Task 3: search box, sort button, Premium label, empty message ===== */
.tools-bar {
  display: flex;
  flex-wrap: wrap;
  gap: 0.75rem;
  padding: 0 1.5rem;
  margin-bottom: 1.5rem;
}

.search-input {
  flex: 1 1 220px;
  max-width: 360px;
  background: #111;
  color: #fff;
  border: 2px solid #444;
  border-radius: 999px;
  padding: 0.45rem 1.1rem;
}
.search-input:focus {
  outline: none;
  border-color: var(--accent);
}

.sort-btn {
  background: transparent;
  color: #fff;
  border: 2px solid var(--accent);
  border-radius: 999px;
  padding: 0.45rem 1.1rem;
  cursor: pointer;
}
.sort-btn:hover {
  background: var(--accent);
}

/* Cards hidden by the search (the category filter hides cards with its own rules) */
.search-hidden {
  display: none;
}

/* Hidden attribute must win over Bootstrap's display:flex on .row */
.product-grid[hidden],
```

```
.no-results[hidden] {
  display: none !important;
}

.no-results {
  text-align: center;
  color: #fff;
  font-size: 1.25rem;
  padding: 3rem 1rem;
}

/* Premium label added by script.js (.card is already position: relative in Bootstrap) */
.premium-badge {
  position: absolute;
  top: 0.5rem;
  left: 0.5rem;
  z-index: 1;
  background: var(--accent);
  color: #fff;
  font-size: 0.75rem;
  font-weight: 700;
  padding: 0.15rem 0.55rem;
  border-radius: 0.25rem;
}
```